Project Report: Custom ASL Detection System

1. Executive Summary

This project outlines the development and deployment of a real-time American Sign Language (ASL) detection system. By leveraging the YOLOv8 object detection architecture, the system provides high-accuracy, low-latency identification of 26 ASL hand signs, effectively bridging a communication gap using computer vision technology.

2. Methodology

2.1 Dataset Creation & Pipeline

- **Custom Data Collection:** The dataset was engineered from the ground up to ensure domain-specific accuracy. Raw images were captured directly from the project environment using a custom `collect_data.py` script, ensuring control over lighting, camera angles, and background variations.
- **Data Splitting:** To ensure objective model evaluation, a custom `data_split.py` script was implemented. This utility shuffles the raw image/label pairs and programmatically distributes them into standard `train`, `val`, and `test` directories (70/15/15 ratio), which is essential for preventing data leakage and ensuring reliable validation.
- **Data Pipeline:** Organized data was processed via a custom `data_pipeline.py` script, which managed resizing, normalization, and label format verification to ensure full compatibility with YOLOv8 requirements.
- **Preprocessing & Augmentation:** Images were resized to 640x640 pixels. Geometric transformations and noise injection were applied to the training set to enhance model robustness under diverse environmental conditions.

2.2 Algorithm & Framework

- **Model Architecture:** YOLOv8 (You Only Look Once), selected for its industry-leading balance between high-speed inference and precision.
- **Software Stack:** Developed using Python 3.13, PyTorch 2.12.0, and the Ultralytics framework.

3. Experimental Setup & Training

- **Hardware:** 13th Gen Intel Core i9-13900H processor.
- **Training Parameters:**
  - **Epochs:** 50
  - **Training Time:** ~1.55 hours
  - **Model Complexity:** 73 layers, ~3.01 million parameters.

4. Results & Discussion

4.1 Quantitative Performance

The model demonstrated high reliability with an overall **mAP@50 of 0.934 (93.4%)** on the test dataset.

4.2 Class Analysis

- **Strengths:** Most signs, including letters A, B, C, and O, achieved exceptional recognition precision.
- **Bottlenecks:** Classes 'I', 'K', and 'N' showed lower mAP scores, indicating minor confusion. These are identified as focal points for future dataset enrichment.

4.3 Inference Speed

The system maintains an inference speed of ~70ms per frame (~14 FPS), which is sufficient for real-time, fluid interaction.

5. Challenges & Solutions

1. **Data Partitioning:** Automated the creation of training/validation/test sets to ensure consistent model evaluation metrics.
2. **Environment Management:** Overcame inconsistencies by standardizing all executions within a dedicated Python virtual environment (`venv`).
3. **Streaming Errors:** Resolved `TypeError` issues in live inference by implementing a generator-based iteration loop (`stream=True`) for real-time video processing.

6. Conclusion & Future Work

6.1 Conclusion

The project confirms that a highly accurate, custom ASL detection system can be developed on consumer-grade hardware. By managing the entire data lifecycle—from custom collection and structured splitting to training and real-time inference—we achieved a functional, low-latency computer vision application.